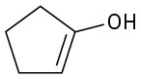
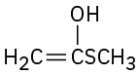
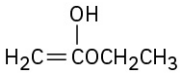
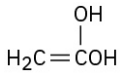
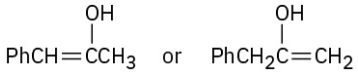
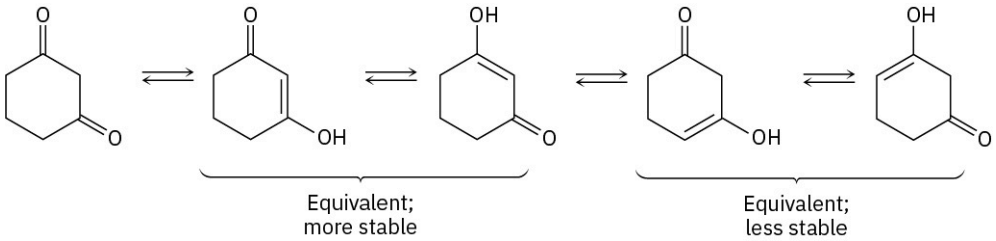
Chapter 22

PROBLEM 22-1 **(a)**  **(b)**  **(c)**  **(d)** $CH_3CH{=}CHOH$ **(e)** 

(f) 

PROBLEM 22-2 **(a)** 4 **(b)** 3 **(c)** 3 **(d)** 2 **(e)** 4 **(f)** 5

PROBLEM 22-3 

PROBLEM 22-4 Acid-catalyzed formation of an enol is followed by deuteronation of the enol double bond and dedeuteronation of oxygen.

PROBLEM 22-5 1. Br_2; 2. Pyridine, heat

PROBLEM 22-6 The intermediate α-bromo acid bromide undergoes a nucleophilic acyl substitution reaction with methanol to give an α-bromo ester.

PROBLEM 22-7 **(a)** CH_3CH_2CHO **(b)** $(CH_3)_3CCOCH_3$ **(c)** CH_3CO_2H **(d)** $PhCONH_2$ **(e)** $CH_3CH_2CH_2CN$ **(f)** $CH_3CON(CH_3)_2$

PROBLEM 22-8 $^{-}{:}CH_2C{\equiv}N{:} \longleftrightarrow H_2C{=}C{=}\ddot{N}{:}^{-}$

PROBLEM 22-9 Acid is regenerated, but base is used stoichiometrically.

PROBLEM 22-10 **(a)** 1. $Na^+\ ^-OEt$; 2. $PhCH_2Br$; 3. H_3O^+
(b) 1. $Na^+\ ^-OEt$; 2. $CH_3CH_2CH_2Br$; 3. $Na^+\ ^-OEt$; 4. CH_3Br; 5. H_3O^+
(c) 1. $Na^+\ ^-OEt$; 2. $(CH_3)_2CHCH_2Br$; 3. H_3O^+

PROBLEM 22-11 Malonic ester has only two acidic hydrogens to be replaced.

PROBLEM 22-12 1. $Na^+\ ^-OEt$; 2. $(CH_3)_2CHCH_2Br$; 3. $Na^+\ ^-OEt$; 4. CH_3Br; 5. H_3O^+

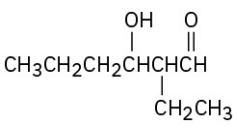
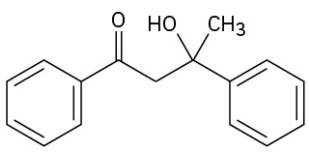
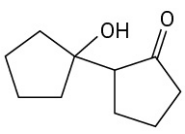
PROBLEM 22-13 **(a)** $(CH_3)_2CHCH_2Br$ **(b)** $PhCH_2CH_2Br$

PROBLEM 22-14 None can be prepared.

PROBLEM 22-15 1. 2 $Na^+\ ^-OEt$; 2. $BrCH_2CH_2CH_2CH_2Br$; 3. H_3O^+

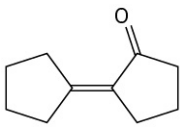
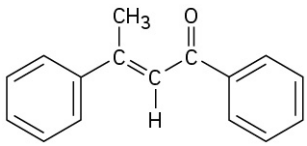
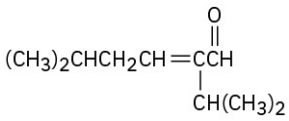
PROBLEM 22-16 **(a)** Alkylate phenylacetone with CH_3I **(b)** Alkylate pentanenitrile with CH_3CH_2I
(c) Alkylate cyclohexanone with $H_2C{=}CHCH_2Br$ **(d)** Alkylate cyclohexanone with excess CH_3I
(e) Alkylate $C_6H_5COCH_2CH_3$ with CH_3I **(f)** Alkylate methyl 3-methylbutanoate with CH_3CH_2I

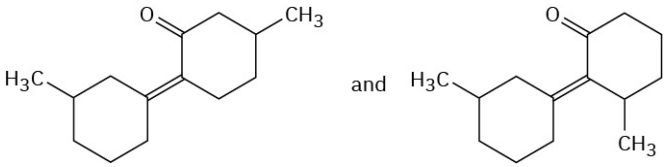
Chapter 23

PROBLEM 23-1 **(a)**  **(b)**  **(c)** 

PROBLEM The reverse reaction is the exact opposite of the forward reaction shown in Figure 23.2.

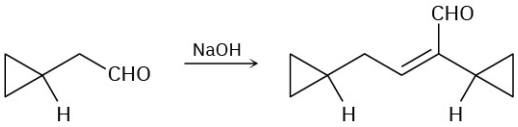
23-2

PROBLEM 23-3 **(a)**  **(b)**  **(c)** 

PROBLEM 23-4 

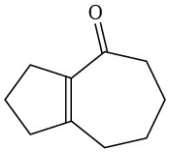
PROBLEM 23-5 **(a)** Not an aldol product **(b)** 3-Pentanone

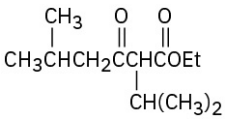
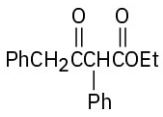
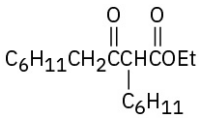
PROBLEM 23-6 1. NaOH; 2. $LiAlH_4$; 3. H_2/Pd

PROBLEM 23-7 

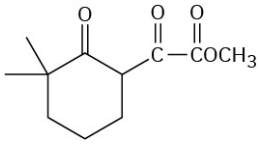
PROBLEM 23-8 **(a)** $C_6H_5CHO + CH_3COCH_3$ **(b)** Not easily prepared **(c)** Not easily prepared

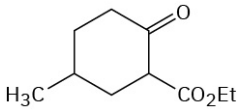
PROBLEM 23-9 The CH_2 position between the two carbonyl groups is so acidic that it is completely deprotonated to give a stable enolate ion.

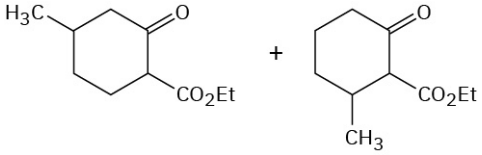
PROBLEM 23-10 

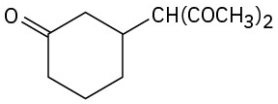
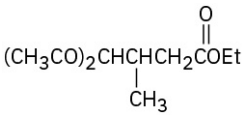
PROBLEM 23-11 **(a)**  **(b)**  **(c)** 

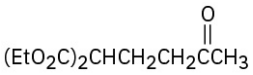
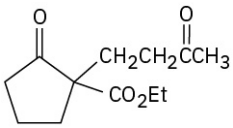
PROBLEM 23-12 The cleavage reaction is the exact reverse of the forward reaction.

PROBLEM 23-13 

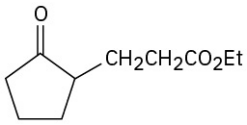
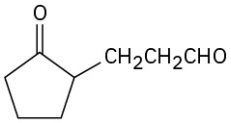
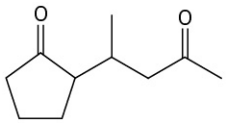
PROBLEM 23-14 

PROBLEM 23-15 

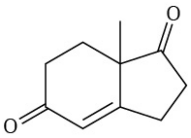
PROBLEM 23-16 **(a)**  **(b)** $(CH_3CO)_2CHCH_2CH_2CN$ **(c)** 

PROBLEM 23-17 **(a)**  **(b)** 

PROBLEM 23-18 $CH_3CH_2COCH{=}CH_2 + CH_3CH_2NO_2$

PROBLEM 23-19 **(a)**  **(b)**  **(c)** 

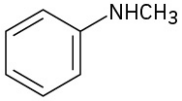
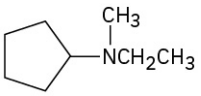
PROBLEM 23-20 **(a)** Cyclopentanone enamine + propenenitrile
(b) Cyclohexanone enamine + methyl propenoate

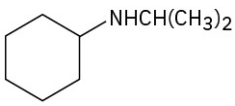
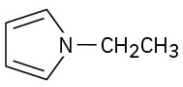
PROBLEM 23-21 

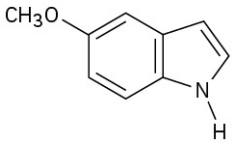
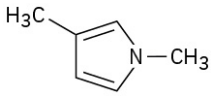
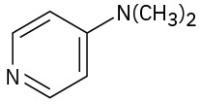
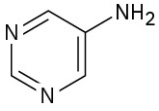
PROBLEM 23-22 2,5,5-Trimethyl-1,3-cyclohexanedione + 1-penten-3-one

Chapter 24

PROBLEM 24-1 **(a)** *N*-Methylethylamine **(b)** Tricyclohexylamine **(c)** *N*-Ethyl-*N*-methylcyclohexylamine **(d)** *N*-Methylpyrrolidine **(e)** Diisopropylamine **(f)** 1,3-Butanediamine

PROBLEM 24-2 **(a)** $[(CH_3)_2CH]_3N$ **(b)** $(H_2C{=}CHCH_2)_3N$ **(c)**  **(d)** 

(e)  **(f)** 

PROBLEM 24-3 **(a)**  **(b)**  **(c)**  **(d)** 

PROBLEM 24-4 **(a)** $CH_3CH_2NH_2$ **(b)** NaOH **(c)** CH_3NHCH_3

PROBLEM 24-5 Propylamine is stronger; benzylamine $pK_b = 4.67$; propylamine $pK_b = 3.29$

PROBLEM 24-6 **(a)** *p*-Nitroaniline < *p*-Aminobenzaldehyde < *p*-Bromoaniline
(b) *p*-Aminoacetophenone < *p*-Chloroaniline < *p*-Methylaniline
(c) *p*-(Trifluoromethyl)aniline < *p*-(Fluoromethyl)aniline < *p*-Methylaniline

PROBLEM 24-7 Pyrimidine is essentially 100% neutral (unprotonated).

PROBLEM 24-8 **(a)** Propanenitrile or propanamide **(b)** *N*-Propylpropanamide **(c)** Benzonitrile or benzamide **(d)** *N*-Phenylacetamide

PROBLEM 24-9 The reaction takes place by two nucleophilic acyl substitution reactions.

PROBLEM 24-10 